Alexander Antonison Data Professional | Consultant

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Over my career, I have performed in Data Engineering and Data Science roles. One consistent focus of mine has been working with external and internal customers to help identify opportunities to more effectively work with data in support of business goals. This has ranged from building automated data pipelines, supporting the construction and deployment of machine learning models, to building out data warehouses to serve analytics platforms.

EXPERIENCE

Antonison Consulting Group, Remote — *Data Solutions Consultant*

June 2020 - Present

At Antonison Consulting Group, I help clients by providing solutions that foster growth and efficiency through Data Engineering and Architecture services.

MTSU Data Science Institute (https://mtsu.edu/datascience/) – November 2021 to Present

Act as an in-house technical expert for building scalable cloud data applications in support of research and education outreach initiatives.

- Building a data platform to take in raw data files into a data lake, process into a data warehouse, and serve up results for users within the Environmental Protection Agency.
    - Built a serverless data lake using AWS S3, Glue, Lambda, and Step Functions for pre-processing the raw sensor and metadata files into a staging location in S3.
    - Used Athena and dbt with Fargate to process the data into a normalized data model.
    - Built data marts to serve data via dashboards in AWS Quicksight.

Concert Genetics (https://www.concertgenetics.com/) – February 2022 to March 2022

Conducted an assessment and provided actionable improvements in the realms of Data Infrastructure, Data Management, and Data Analytics workflow.

Verana Health (https://www.veranahealth.com/) – November 2021 to March 2022

Provided support as a Data Solutions Consultant to Verana Health as they are building a data lake for ingesting thousands of Electronic Health Record (EHR) systems into Real World Evidence data repositories.

Picknic (https://picknic.app/) – June 2020

Reviewed client's existing data ingestion and architecture and provided guidance on industry best practices to streamline the ingestion and management of data within the Picknic application.

Verana Health, Remote — *Data Engineer*

March 2021 - November 2021

At Verana Health, I worked on building an ingestion platform that pulls in data from Electronic Health Record (EHR) systems and processes them into a common data model for further processing and analysis.

- Built PySpark Data Quality modules that profile and evaluate data as it flows through Ingestion Pipeline using AWS's Deequ library.
- Built AWS Glue PySpark jobs to ingest incremental data pulls into a staged data layer orchestrated by AWS Step Functions.
- Developed application templates using cookiecutter to provide a standardized method of creating AWS Glue Jobs and Serverless Applications. Templates included code for unit testing, generating fake data, and CI/CD configuration files for GitHub Actions.

SKILLS

Programming

R, Python, SQL, BASH, Markdown

Data Technology

MSSQL, Postgres, Hive, Presto, Spark, Salesforce, Snowflake

Cloud Technology

AWS: Lambda, S3, EC2, Glue, Step Functions, Athena, Redshift, EMR, IAM, Quicksight, ECR, Fargate, CloudWatch

Google Cloud: BigQuery, Google Cloud Storage

Dev Technology

Serverless, Docker, GitHub, Bitbucket, PyTest, GitHub Actions, Cookiecutter

Engineering Tools

dbt, Deequ, Talend, Tableau, PowerBI, SSRS, Excel(PowerQuery/PowerPivot), RStudio Server Pro, JupyterHub, RMarkdown

Stratasan, Nashville, TN — *Principal Machine Learning Engineer*

October 2019 - February 2021

I started at Stratasan as the Data Services Manager where I lead the Data Services team that consisted of Data Engineering and Data Science. I worked to cast the technical vision for data management and machine learning as well as manage, develop, and grow the team. I transitioned to Principal Machine Learning Engineer to focus on identifying ways to help Stratasan scale data products and support the development and deployment of machine learning models.

- Established a source control strategy and standardized development processes and environment to help ensure quality and efficiency.
- Developed and mentored Data Engineers and Data Scientists.
- Worked closely with the Product team to provide technical guidance on enhancing and developing new data products.
- Implemented reference data management process to improve control and communication.
- Provided technical guidance on how to scale data products. This involved identifying new technologies, optimizing data, and automating manual data processes.
  - Optimized data warehouse in Redshift cluster through rewriting SQL queries, introducing sort and dist keys, and optimizing data types for filter and relation columns.
  - Introduced dbt as a streamlined method for managing Extract Load Transform (ELT) pipelines.
  - Recommended S3 Lifecycle policies to optimize S3 spend.

Axial Healthcare, Nashville, TN — *Data Engineer -> Data Scientist*

July 2017 - September 2019

I worked on the Data Operations team to help enhance and manage the intake and processing of client and reference data. I transferred to the Data Science team where my focus was to streamline the development, deployment, and management of models.

- Improved model and analysis development efficiency by developing and maintaining inhouse R and python packages to streamline development actions and feature development.
- Streamlined the deployment of models to production through the use of virtual environments, docker, and adding error handling, logging, and unit testing to model code.
- Provided development guidance related to coding practices and source control to fellow data scientists and researchers.
- Reduced development downtime by ensuring compute resources for the data science team were operating nominally. This included RStudio Server Pro, JupyterHub, and Redshift.
- Ingested and managed data stored in Redshift in support of Machine Learning research. This involved creating optimized versions of tables in support of different research projects.
- Contributed to the development of an enhanced metric engine by coding the data extraction layer in Scala with Spark.

Education

M.C.S Data Science
University of Illinois at Urbana-Champaign

August 2017 - August 2020

B.E. Industrial and Systems Engineering
University of Alabama in Huntsville

May 2007 - December 2011

FreseniusRx, Nashville, TN — *Data Engineer/Analyst*

April 2015 - July 2017

Supported analytic groups within Fresenius and FreseniusRx along with day to day operations through developing data pipelines, warehousing data, developing and maintaining reporting and dashboard solutions.

- Improved the quality and speed of delivering data and reports to the business by developing scheduled data pipelines using Talend that processed application and third party data into a data warehouse.
- Increased users ability to access and utilize data by curating data marts and supporting self-service analytics with PowerBI and Excel (PowerQuery and PowerPivot)
- Improved visibility into operational and compliance issues by developing automated report alerts that e-mailed stakeholders using SSRS and Talend.
- Streamlined the extraction and loading of data into Salesforce using Talend in support of call center operations.

Oilfield Instrumentation, Lafayette, LA — *System Analyst*

August 2014 - March 2015

Worked in the engineering department of an oilfield drilling instrumentation company and was responsible for supporting development, testing, and troubleshooting field issues for multiple OI software products.

- Improved the maintainability and efficiency of the customer SSRS report suite by optimizing the underlying SQL.
- Supported software development by documenting and replicating software bugs or feature issues. Would additionally collect changes to business logic if necessary.
- Interacted with customers to assess possible software issues and provide technical support.

Wyle, CAS, Huntsville, AL — *System Analyst*

May 2012 - August 2014

Worked as an analyst in support of PATRIOT System Effectiveness Model (PSEM), a simulation model of the PATRIOT missile defense system.

- Reduced the time of producing a comprehensive evaluation of PSEM's performance in integrated test events from days to a few hours. Accomplished this with a series of shell scripts that loaded data into the database to queries that processed and analyzed the results.
- Helped improve software development efficiency by developing BASH shell scripts that automated running software test suite and simulation test cases.
- Performed Verification and Validation activities on PSEM, co-authored reports on the findings, and presented results to PATRIOT project office.